Cumulant expansion of the three-body terms

generated by `cumulant_expansion.py`

The three-body operators surviving the space restriction are decomposed into two-body and one-body operators (plus scalar terms) with density-matrix coefficients, using the spin-free cumulant decomposition of `cumulant_decomposition/spin_free/spin_free.tex`:

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx = {} & D^p_s \hat{E}^{qr}_{tu} - \tfrac{1}{2}\, D^p_t \hat{E}^{qr}_{su} - \tfrac{1}{2}\, D^p_u \hat{E}^{qr}_{ts} && (1)\\
& - \tfrac{1}{2}\, D^q_s \hat{E}^{pr}_{tu} + D^q_t \hat{E}^{pr}_{su} - \tfrac{1}{2}\, D^q_u \hat{E}^{pr}_{st} && (2)\\
& - \tfrac{1}{2}\, D^r_s \hat{E}^{qp}_{tu} - \tfrac{1}{2}\, D^r_t \hat{E}^{pq}_{su} + D^r_u \hat{E}^{pq}_{st} + D^{pq}_{st} \hat{E}^r_u && (3)\\
& - \tfrac{1}{2}\, D^{pq}_{su} \hat{E}^r_t - \tfrac{1}{2}\, D^{pq}_{ut} \hat{E}^r_s - \tfrac{1}{2}\, D^{pr}_{st} \hat{E}^q_u && (4)\\
& + D^{pr}_{su} \hat{E}^q_t - \tfrac{1}{2}\, D^{pr}_{tu} \hat{E}^q_s - \tfrac{1}{2}\, D^{qr}_{su} \hat{E}^p_t + D^{qr}_{tu} \hat{E}^p_s && (5)\\
& - \tfrac{1}{2}\, D^{rq}_{st} \hat{E}^p_u - 2\, D^p_s D^q_t \hat{E}^r_u - 2\, D^p_s D^r_u \hat{E}^q_t && (6)\\
& - \tfrac{1}{2}\, D^p_t D^q_u \hat{E}^r_s - \tfrac{1}{2}\, D^p_t D^r_s \hat{E}^q_u - \tfrac{1}{2}\, D^p_u D^q_s \hat{E}^r_t && (7)\\
& - \tfrac{1}{2}\, D^p_u D^r_t \hat{E}^q_s - \tfrac{1}{2}\, D^q_s D^r_t \hat{E}^p_u - 2\, D^q_t D^r_u \hat{E}^p_s && (8)\\
& - \tfrac{1}{2}\, D^q_u D^r_s \hat{E}^p_t + D^p_s D^q_u \hat{E}^r_t + D^p_s D^r_t \hat{E}^q_u && (9)\\
& + D^p_t D^q_s \hat{E}^r_u + D^p_t D^r_u \hat{E}^q_s + D^p_u D^q_t \hat{E}^r_s + D^p_u D^r_s \hat{E}^q_t && (10)\\
& + D^q_s D^r_u \hat{E}^p_t + D^q_t D^r_s \hat{E}^p_u + D^q_u D^r_t \hat{E}^p_s - D^p_s D^{qr}_{tu} && (11)\\
& + \tfrac{1}{2}\, D^p_t D^{qr}_{su} + \tfrac{1}{2}\, D^p_u D^{rq}_{st} + \tfrac{1}{2}\, D^q_s D^{pr}_{tu} - D^q_t D^{pr}_{su} && (12)\\
& + \tfrac{1}{2}\, D^q_u D^{pr}_{st} + \tfrac{1}{2}\, D^r_s D^{pq}_{ut} + \tfrac{1}{2}\, D^r_t D^{pq}_{su} - D^r_u D^{pq}_{st} && (13)\\
& + 4\, D^p_s D^q_t D^r_u - 2\, D^p_s D^q_u D^r_t - 2\, D^p_t D^q_s D^r_u + D^p_t D^q_u D^r_s && (14)\\
& + D^p_u D^q_s D^r_t - 2\, D^p_u D^q_t D^r_s && (15)
\end{aligned}$$

This is an operator identity: the labels p, q, r, s, t, u are fixed on both sides, so no index may be renamed; only the index pairs of each operator can be permuted together (the diagonal pair-permutation symmetry of the spin-summed $\hat{E}$, no sign). With the terms grouped by the operator in this sense, the decomposition reads

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx {} & \hat{E}^{pq}_{st}\ (D^r_u) && (16)\\
& \hat{E}^{pq}_{su}\ \left(-\tfrac{1}{2}\, D^r_t\right) && (17)\\
& \hat{E}^{pq}_{ut}\ \left(-\tfrac{1}{2}\, D^r_s\right) && (18)\\
& \hat{E}^{pr}_{st}\ \left(-\tfrac{1}{2}\, D^q_u\right) && (19)\\
& \hat{E}^{pr}_{su}\ (D^q_t) && (20)\\
& \hat{E}^{pr}_{tu}\ \left(-\tfrac{1}{2}\, D^q_s\right) && (21)\\
& \hat{E}^{qr}_{su}\ \left(-\tfrac{1}{2}\, D^p_t\right) && (22)
\end{aligned}$$

$$\hat{E}^{qr}_{ts}\ \left(-\tfrac{1}{2}\,D^{p}_{u}\right) \tag{23}$$
$$\hat{E}^{qr}_{tu}\ \left(D^{p}_{s}\right) \tag{24}$$
$$\hat{E}^{p}_{s}\ \left(D^{qr}_{tu} - 2\,D^{q}_{t}D^{r}_{u} + D^{q}_{u}D^{r}_{t}\right) \tag{25}$$
$$\hat{E}^{p}_{t}\ \left(-\tfrac{1}{2}\,D^{qr}_{su} + D^{q}_{s}D^{r}_{u} - \tfrac{1}{2}\,D^{q}_{u}D^{r}_{s}\right) \tag{26}$$
$$\hat{E}^{p}_{u}\ \left(-\tfrac{1}{2}\,D^{q}_{s}D^{r}_{t} + D^{q}_{t}D^{r}_{s} - \tfrac{1}{2}\,D^{rq}_{st}\right) \tag{27}$$
$$\hat{E}^{q}_{s}\ \left(-\tfrac{1}{2}\,D^{pr}_{tu} + D^{p}_{t}D^{r}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{r}_{t}\right) \tag{28}$$
$$\hat{E}^{q}_{t}\ \left(D^{pr}_{su} - 2\,D^{p}_{s}D^{r}_{u} + D^{p}_{u}D^{r}_{s}\right) \tag{29}$$
$$\hat{E}^{q}_{u}\ \left(-\tfrac{1}{2}\,D^{pr}_{st} + D^{p}_{s}D^{r}_{t} - \tfrac{1}{2}\,D^{p}_{t}D^{r}_{s}\right) \tag{30}$$
$$\hat{E}^{r}_{s}\ \left(-\tfrac{1}{2}\,D^{pq}_{ut} - \tfrac{1}{2}\,D^{p}_{t}D^{q}_{u} + D^{p}_{u}D^{q}_{t}\right) \tag{31}$$
$$\hat{E}^{r}_{t}\ \left(-\tfrac{1}{2}\,D^{pq}_{su} + D^{p}_{s}D^{q}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{q}_{s}\right) \tag{32}$$
$$\hat{E}^{r}_{u}\ \left(D^{pq}_{st} - 2\,D^{p}_{s}D^{q}_{t} + D^{p}_{t}D^{q}_{s}\right) \tag{33}$$
$$- D^{p}_{s}D^{qr}_{tu} + 4\,D^{p}_{s}D^{q}_{t}D^{r}_{u} - 2\,D^{p}_{s}D^{q}_{u}D^{r}_{t} + \tfrac{1}{2}\,D^{p}_{t}D^{qr}_{su} \tag{34}$$
$$- 2\,D^{p}_{t}D^{q}_{s}D^{r}_{u} + D^{p}_{t}D^{q}_{u}D^{r}_{s} + D^{p}_{u}D^{q}_{s}D^{r}_{t} \tag{35}$$
$$- 2\,D^{p}_{u}D^{q}_{t}D^{r}_{s} + \tfrac{1}{2}\,D^{p}_{u}D^{rq}_{st} + \tfrac{1}{2}\,D^{q}_{s}D^{pr}_{tu} \tag{36}$$
$$- D^{q}_{t}D^{pr}_{su} + \tfrac{1}{2}\,D^{q}_{u}D^{pr}_{st} + \tfrac{1}{2}\,D^{r}_{s}D^{pq}_{ut} \tag{37}$$
$$+ \tfrac{1}{2}\,D^{r}_{t}D^{pq}_{su} - D^{r}_{u}D^{pq}_{st} \tag{38}$$

(19 operator groups; a group with no $\hat{E}$ shown is the scalar part.) The $\approx$ drops the normal-ordered three-body remainder and the three-cumulant λ^{pqr}_{stu}; for a single-determinant reference $\lambda_3 = 0$ identically. Every density matrix carrying a virtual index vanishes; density matrices with occupied or whole-space indices are kept symbolic as D^{i}_{j}, D^{ij}_{kl}, … (no delta-function substitution). The formula is verified numerically for a closed-shell single-determinant reference on Jordan–Wigner matrices (random index assignments with and without the virtual rule).

1 Final blocks

Only the blocks containing surviving three-body terms change; the three-body pieces are decomposed and merged back into the block. Blocks not shown are unchanged from `space_restriction.tex`.

$\left[\hat{V},\,\hat{T}_2\right]$

after the space restriction (6 terms, 4 three-body):

$$\left[\hat{V},\,\hat{T}_2\right] = \tfrac{1}{4}\,g^{a_1a_2}_{p_1p_2}t^{a_1a_2}_{i_1i_2}\,\hat{E}^{p_1p_2}_{i_1i_2} \tag{39}$$
$$+ \tfrac{1}{4}\,g^{a_1a_2}_{p_2p_1}t^{a_1a_2}_{i_2i_1}\,\hat{E}^{p_1p_2}_{i_1i_2} \tag{40}$$
$$+ \tfrac{1}{4}\,g^{a_2p_3}_{p_1p_2}t^{a_1a_2}_{i_1i_2}\,\hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \tag{41}$$
$$+ \tfrac{1}{4}\,g^{a_2p_3}_{p_1p_2}t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \tag{42}$$
$$+ \tfrac{1}{4}\,g^{a_2p_3}_{p_2p_1}t^{a_1a_2}_{i_1i_2}\,\hat{E}^{a_1p_1p_2}_{i_1p_3i_2} \tag{43}$$
$$+ \tfrac{1}{4}\,g^{a_2p_3}_{p_2p_1}t^{a_2a_1}_{i_2i_1}\,\hat{E}^{a_1p_1p_2}_{i_1p_3i_2} \tag{44}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives

iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \approx \hat{E}^{a1p1}_{i1i2} \ \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2}_{p_3}\right) \tag{45}$$

$$\hat{E}^{a1p1}_{i1p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_2}\right) \tag{46}$$

$$\hat{E}^{a1p1}_{p3i2} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_1}\right) \tag{47}$$

$$\hat{E}^{a1p2}_{i1i2} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3}\right) \tag{48}$$

$$\hat{E}^{a1p2}_{i1p3} \ \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2}\right) \tag{49}$$

$$\hat{E}^{a1p2}_{i2p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1}\right) \tag{50}$$

$$\hat{E}^{a1}_{i1} \ \Big(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{i_2p_3} \tag{51}$$

$$-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{p_3} \tag{52}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_2}\Big) \tag{53}$$

$$\hat{E}^{a1}_{i2} \ \Big(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{i_1p_3} \tag{54}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{p_3} \tag{55}$$

$$-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_1}\Big) \tag{56}$$

$$\hat{E}^{a1}_{p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{i_2}\right. \tag{57}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{i_1} \tag{58}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_2a_1}_{i_2i_1} D^{p_2p_1}_{i_1i_2}\right) \tag{59}$$

9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1p_2}_{i_1i_2p_3} \approx \hat{E}^{a1p1}_{i1i2} \ \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_2}_{p_3}\right) \tag{60}$$

$$\hat{E}^{a1p1}_{i1p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_2}_{i_2}\right) \tag{61}$$

$$\hat{E}^{a1p1}_{p3i2} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_2}_{i_1}\right) \tag{62}$$

$$\hat{E}^{a1p2}_{i1i2} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3}\right) \tag{63}$$

$$\hat{E}^{a1p2}_{i1p3} \ \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2}\right) \tag{64}$$

$$\hat{E}^{a1p2}_{i2p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1}\right) \tag{65}$$

$$\hat{E}^{a1}_{i1} \ \Big(\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1p_2}_{i_2p_3} \tag{66}$$

$$-\tfrac{1}{2}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2} D^{p_2}_{p_3} \tag{67}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3} D^{p_2}_{i_2}\Big) \tag{68}$$

$$\hat{E}^{a1}_{i2} \ \Big(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1p_2}_{i_1p_3} \tag{69}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1} D^{p_2}_{p_3} \tag{70}$$

$$-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3} D^{p_2}_{i_1}\Big) \tag{71}$$

$$\hat{E}^{a1}_{p3} \ \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1} D^{p_2}_{i_2}\right. \tag{72}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2} D^{p_2}_{i_1} \tag{73}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} D^{p_2p_1}_{i_1i_2}\right) \tag{74}$$

9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} \hat{E}^{a_1p_1p_2}_{i_1p_3i_2} \approx \hat{E}^{a1p1}_{i1p3} \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_2}\right) \tag{75}$$

$$\hat{E}^{a1p1}_{i1i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_2}_{p_3}\right) \tag{76}$$

$$\hat{E}^{a1p1}_{i2p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_2}_{i_1}\right) \tag{77}$$

$$\hat{E}^{a1p2}_{i1p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2}\right) \tag{78}$$

$$\hat{E}^{a1p2}_{i1i2} \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3}\right) \tag{79}$$

$$\hat{E}^{a1p2}_{p3i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1}\right) \tag{80}$$

$$\hat{E}^{a1}_{i1} \Big(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{p_3i_2} \tag{81}$$

$$-\tfrac{1}{2}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_2} \tag{82}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{p_3}\Big) \tag{83}$$

$$\hat{E}^{a1}_{p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1p_2}_{i_1i_2} \right. \tag{84}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{i_2} \tag{85}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_2} D^{p_2}_{i_1}\right) \tag{86}$$

$$\hat{E}^{a1}_{i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{i_1} D^{p_2}_{p_3} \right. \tag{87}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_1}_{p_3} D^{p_2}_{i_1} \tag{88}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_2a_1}_{i_2i_1} D^{p_2p_1}_{i_1p_3}\right) \tag{89}$$

9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1p_2}_{i_1p_3i_2} \approx \hat{E}^{a1p1}_{i1p3} \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_2}_{i_2}\right) \tag{90}$$

$$\hat{E}^{a1p1}_{i1i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_2}_{p_3}\right) \tag{91}$$

$$\hat{E}^{a1p1}_{i2p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_2}_{i_1}\right) \tag{92}$$

$$\hat{E}^{a1p2}_{i1p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2}\right) \tag{93}$$

$$\hat{E}^{a1p2}_{i1i2} \left(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3}\right) \tag{94}$$

$$\hat{E}^{a1p2}_{p3i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1}\right) \tag{95}$$

$$\hat{E}^{a1}_{i1} \Big(\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1p_2}_{p_3i_2} \tag{96}$$

$$-\tfrac{1}{2}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3} D^{p_2}_{i_2} \tag{97}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2} D^{p_2}_{p_3}\Big) \tag{98}$$

$$\hat{E}^{a1}_{p3} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1p_2}_{i_1i_2} \right. \tag{99}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1} D^{p_2}_{i_2} \tag{100}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_2} D^{p_2}_{i_1}\right) \tag{101}$$

$$\hat{E}^{a1}_{i2} \left(-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{i_1} D^{p_2}_{p_3} \right. \tag{102}$$

$$+\tfrac{1}{4}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_1}_{p_3} D^{p_2}_{i_1} \tag{103}$$

$$\left.-\tfrac{1}{8}\, g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} D^{p_2p_1}_{i_1p_3}\right) \tag{104}$$

the decomposition of the three-body part produces 15 terms; the final block (surviving terms and decomposition merged) has 17 terms:

$$\begin{aligned}
\left[\hat{V},\,\hat{T}_2\right] &= D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{i_1} && (105)\\
&- \tfrac{1}{2}\, D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_2i_1} \hat{E}^{a_1}_{i_1} && (106)\\
&- 2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{i_1} && (107)\\
&+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_2i_1} \hat{E}^{a_1}_{i_1} && (108)\\
&+ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{i_1} && (109)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_2i_1} \hat{E}^{a_1}_{i_1} && (110)\\
&- \tfrac{1}{2}\, D^{i_1i_2}_{p_2p_3} g^{a_2p_1}_{p_3p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{p_1} && (111)\\
&- \tfrac{1}{2}\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{p_1} && (112)\\
&+ D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_3p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1}_{p_1} && (113)\\
&+ D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1}_{i_1i_2} && (114)\\
&- \tfrac{1}{2}\, D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1}_{i_1i_2} && (115)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1}_{i_1p_2} && (116)\\
&+ D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_1a_2}_{i_1i_2} \hat{E}^{a_1p_1}_{i_1p_2} && (117)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_3p_1} t^{a_1a_2}_{i_2i_1} \hat{E}^{a_1p_1}_{i_1p_2} && (118)\\
&- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_1a_2}_{i_2i_1} \hat{E}^{a_1p_1}_{p_2i_1} && (119)\\
&+ \tfrac{1}{4}\, g^{a_1a_2}_{p_1p_2} t^{a_1a_2}_{i_1i_2} \hat{E}^{p_1p_2}_{i_1i_2} && (120)\\
&+ \tfrac{1}{4}\, g^{a_1a_2}_{p_2p_1} t^{a_1a_2}_{i_2i_1} \hat{E}^{p_1p_2}_{i_1i_2} && (121)
\end{aligned}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (6 operator groups):

$$\begin{aligned}
\left[\hat{V},\,\hat{T}_2\right] &= \hat{E}^{a1}_{i1}\ \big(D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_1i_2} && (122)\\
&\quad - \tfrac{1}{2}\, D^{i_2p_1}_{p_2p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_2i_1} && (123)\\
&\quad - 2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} && (124)\\
&\quad + D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_2i_1} && (125)\\
&\quad + D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} && (126)\\
&\quad - \tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_2i_1} \big) && (127)\\
&\hat{E}^{a1}_{p1}\ \big(- \tfrac{1}{2}\, D^{i_1i_2}_{p_2p_3} g^{a_2p_1}_{p_3p_2} t^{a_1a_2}_{i_1i_2} && (128)\\
&\quad - \tfrac{1}{2}\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_2p_3} t^{a_1a_2}_{i_1i_2} && (129)\\
&\quad + D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2p_1}_{p_3p_2} t^{a_1a_2}_{i_1i_2} \big) && (130)\\
&\hat{E}^{p1a1}_{i1p2}\ \big(- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_1a_2}_{i_2i_1} \big) && (131)\\
&\hat{E}^{p1a1}_{i2i1}\ \big(D^{p_2}_{p_3} g^{a_2p_3}_{p_1p_2} t^{a_1a_2}_{i_1i_2} && (132)\\
&\quad - \tfrac{1}{2}\, D^{p_2}_{p_3} g^{a_2p_3}_{p_2p_1} t^{a_1a_2}_{i_1i_2} \big) && (133)\\
&\hat{E}^{p1a1}_{p2i1}\ \big(- \tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2p_2}_{p_1p_3} t^{a_1a_2}_{i_1i_2} && (134)
\end{aligned}$$

$$+D^{i_2}_{p_3}\, g^{a_2 p_2}_{p_3 p_1}\, t^{a_1 a_2}_{i_1 i_2} \tag{135}$$
$$-\tfrac{1}{2}\, D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1}\, t^{a_1 a_2}_{i_2 i_1}\big) \tag{136}$$
$$\hat{E}^{p1p2}_{i1i2}\ \big(\tfrac{1}{4}\, g^{a_1 a_2}_{p_1 p_2}\, t^{a_1 a_2}_{i_1 i_2} \tag{137}$$
$$+\tfrac{1}{4}\, g^{a_1 a_2}_{p_2 p_1}\, t^{a_1 a_2}_{i_2 i_1}\big) \tag{138}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^{\dagger},\, \hat{T}_2\right]$$

after the space restriction (14 terms, 10 three-body):

$$\left[\left[\hat{F},\, \hat{T}_2\right]^{\dagger},\, \hat{T}_2\right] = \tfrac{1}{2}\, f^{a_1}_{a_2}\, t^{a_1 a_3}_{i_1 i_2}\, t^{a_3 a_2}_{i_4 i_3}\, \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{139}$$
$$+\ \tfrac{1}{2}\, f^{a_1}_{a_2}\, t^{a_3 a_1}_{i_1 i_2}\, t^{a_3 a_2}_{i_3 i_4}\, \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{140}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_1 a_2}_{i_1 i_4}\, t^{a_1 a_2}_{i_2 i_3}\, \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{141}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_1 a_2}_{i_4 i_1}\, t^{a_1 a_2}_{i_3 i_2}\, \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{142}$$
$$+\ \tfrac{1}{4}\, f^{a_3}_{a_4}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_1 a_4}_{i_3 i_4}\, \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{143}$$
$$+\ \tfrac{1}{4}\, f^{a_3}_{a_4}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_4 a_1}_{i_4 i_3}\, \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{144}$$
$$+\ \tfrac{1}{4}\, f^{a_3}_{a_4}\, t^{a_3 a_2}_{i_1 i_2}\, t^{a_1 a_4}_{i_3 i_4}\, \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 a_2} \tag{145}$$
$$+\ \tfrac{1}{4}\, f^{a_3}_{a_4}\, t^{a_3 a_2}_{i_1 i_2}\, t^{a_4 a_1}_{i_4 i_3}\, \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 a_2} \tag{146}$$
$$+\ \tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_3 a_2}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{147}$$
$$+\ \tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, \hat{E}^{a_1 i_1 i_2}_{i_3 p_1 i_4} \tag{148}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_2 a_3}_{i_1 i_4}\, t^{a_3 a_1}_{i_3 i_2}\, \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{149}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_3 a_2}_{i_4 i_1}\, t^{a_3 a_1}_{i_3 i_2}\, \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{150}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_2 a_3}_{i_4 i_1}\, t^{a_3 a_1}_{i_3 i_2}\, \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \tag{151}$$
$$-\ \tfrac{1}{2}\, f^{i_4}_{p_1}\, t^{a_3 a_2}_{i_1 i_4}\, t^{a_3 a_1}_{i_3 i_2}\, \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \tag{152}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, \hat{E}^{a_1 i_1 i_2}_{i_3 p_1 i_4} \approx \hat{E}^{a1i1}_{i3p1}\ \left(\tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_2}_{i_4}\right) \tag{153}$$
$$\hat{E}^{a1i1}_{i3i4}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_2}_{p_1}\right) \tag{154}$$
$$\hat{E}^{a1i1}_{i4p1}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_2}_{i_3}\right) \tag{155}$$
$$\hat{E}^{a1i2}_{i3p1}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_1}_{i_4}\right) \tag{156}$$
$$\hat{E}^{a1i2}_{i3i4}\ \left(\tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_1}_{p_1}\right) \tag{157}$$
$$\hat{E}^{a1i2}_{p1i4}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_1}_{i_3}\right) \tag{158}$$
$$\hat{E}^{a1}_{i3}\ \left(\tfrac{1}{2}\, f^{a_2}_{p_1}\, t^{a_2 a_3}_{i_1 i_2}\, t^{a_3 a_1}_{i_4 i_3}\, D^{i_1 i_2}_{p_1 i_4}\right. \tag{159}$$

$$-f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{p_1} D^{i_2}_{i_4} \tag{160}$$

$$+\tfrac{1}{2} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{i_4} D^{i_2}_{p_1}\Big) \tag{161}$$

$$\hat{E}^{a1}_{p1} \Big(-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1 i_2}_{i_3 i_4} \tag{162}$$

$$+\tfrac{1}{2} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{163}$$

$$-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{i_4} D^{i_2}_{i_3}\Big) \tag{164}$$

$$\hat{E}^{a1}_{i4} \Big(-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{i_3} D^{i_2}_{p_1} \tag{165}$$

$$+\tfrac{1}{2} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{p_1} D^{i_2}_{i_3} \tag{166}$$

$$-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_2 i_1}_{i_3 p_1}\Big) \tag{167}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 a_2} \approx \hat{E}^{a1i1}_{i3a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_2}_{i_4}\Big) \tag{168}$$

$$\hat{E}^{a1i1}_{a2i4} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_2}_{i_3}\Big) \tag{169}$$

$$\hat{E}^{a1i2}_{i3a2} \Big(\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_1}_{i_4}\Big) \tag{170}$$

$$\hat{E}^{a1i2}_{i4a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_1}_{i_3}\Big) \tag{171}$$

$$\hat{E}^{a1}_{a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{172}$$

$$+\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_1}_{i_4} D^{i_2}_{i_3} \tag{173}$$

$$-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_4 a_1}_{i_4 i_3} D^{i_2 i_1}_{i_3 i_4}\Big) \tag{174}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 a_2} \approx \hat{E}^{a1i1}_{i3a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_2}_{i_4}\Big) \tag{175}$$

$$\hat{E}^{a1i1}_{a2i4} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_2}_{i_3}\Big) \tag{176}$$

$$\hat{E}^{a1i2}_{i3a2} \Big(\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_1}_{i_4}\Big) \tag{177}$$

$$\hat{E}^{a1i2}_{i4a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_1}_{i_3}\Big) \tag{178}$$

$$\hat{E}^{a1}_{a2} \Big(-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{179}$$

$$+\tfrac{1}{4} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_1}_{i_4} D^{i_2}_{i_3} \tag{180}$$

$$-\tfrac{1}{8} f^{a_3}_{a_4} t^{a_3 a_2}_{i_1 i_2} t^{a_1 a_4}_{i_3 i_4} D^{i_2 i_1}_{i_3 i_4}\Big) \tag{181}$$

9 of the 19 groups survive (6 two-body, 3 one-body):

$$\tfrac{1}{2} f^{a_2}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \approx \hat{E}^{a1i1}_{i3i4} \Big(\tfrac{1}{2} f^{a_2}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_2}_{p_1}\Big) \tag{182}$$

$$\hat{E}^{a1i1}_{i3p1} \Big(-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_2}_{i_4}\Big) \tag{183}$$

$$\hat{E}^{a1i1}_{p1i4} \Big(-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_2}_{i_3}\Big) \tag{184}$$

$$\hat{E}^{a1i2}_{i3i4} \Big(-\tfrac{1}{4} f^{a_2}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_3 a_1}_{i_4 i_3} D^{i_1}_{p_1}\Big) \tag{185}$$

$$\hat{E}^{a1i2}_{i3p1}\ \left(\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4}\right) \tag{186}$$

$$\hat{E}^{a1i2}_{i4p1}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3}\right) \tag{187}$$

$$\hat{E}^{a1}_{i3}\ \left(\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1i_2}_{i_4p_1}\right. \tag{188}$$

$$-f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4} D^{i_2}_{p_1} \tag{189}$$

$$\left.+\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{p_1} D^{i_2}_{i_4}\right) \tag{190}$$

$$\hat{E}^{a1}_{i4}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1i_2}_{i_3p_1}\right. \tag{191}$$

$$+\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3} D^{i_2}_{p_1} \tag{192}$$

$$\left.-\tfrac{1}{4}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{p_1} D^{i_2}_{i_3}\right) \tag{193}$$

$$\hat{E}^{a1}_{p1}\ \left(-\tfrac{1}{4}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_3} D^{i_2}_{i_4}\right. \tag{194}$$

$$+\tfrac{1}{2}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_1}_{i_4} D^{i_2}_{i_3} \tag{195}$$

$$\left.-\tfrac{1}{4}\, f^{a_2}_{p_1} t^{a_3a_2}_{i_1i_2} t^{a_3a_1}_{i_4i_3} D^{i_2i_1}_{i_3i_4}\right) \tag{196}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} \hat{E}^{a_1i_1i_2}_{i_3a_2i_4} \approx \hat{E}^{a1i1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_2}_{i_4}\right) \tag{197}$$

$$\hat{E}^{a1i1}_{i4a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_2}_{i_3}\right) \tag{198}$$

$$\hat{E}^{a1i2}_{i3a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_1}_{i_4}\right) \tag{199}$$

$$\hat{E}^{a1i2}_{a2i4}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_1}_{i_3}\right) \tag{200}$$

$$\hat{E}^{a1}_{a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_1i_2}_{i_3i_4}\right. \tag{201}$$

$$+\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{202}$$

$$\left.-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_4a_1}_{i_4i_3} D^{i_1}_{i_4} D^{i_2}_{i_3}\right) \tag{203}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} \hat{E}^{a_1i_1i_2}_{i_3a_2i_4} \approx \hat{E}^{a1i1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_2}_{i_4}\right) \tag{204}$$

$$\hat{E}^{a1i1}_{i4a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_2}_{i_3}\right) \tag{205}$$

$$\hat{E}^{a1i2}_{i3a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_1}_{i_4}\right) \tag{206}$$

$$\hat{E}^{a1i2}_{a2i4}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_1}_{i_3}\right) \tag{207}$$

$$\hat{E}^{a1}_{a2}\ \left(-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_1i_2}_{i_3i_4}\right. \tag{208}$$

$$+\tfrac{1}{4}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_1}_{i_3} D^{i_2}_{i_4} \tag{209}$$

$$\left.-\tfrac{1}{8}\, f^{a_3}_{a_4} t^{a_2a_3}_{i_1i_2} t^{a_1a_4}_{i_3i_4} D^{i_1}_{i_4} D^{i_2}_{i_3}\right) \tag{210}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} \hat{E}^{a_1i_1p_1}_{i_2a_2i_3} \approx \hat{E}^{a1i1}_{i2a2}\ \left(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_3}\right) \tag{211}$$

$$\hat{E}^{a1i1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_2}\right) \tag{212}$$

$$\hat{E}^{a1p1}_{i2a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3}\right) \tag{213}$$

$$\hat{E}^{a1p1}_{a2i3}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2}\right) \tag{214}$$

$$\hat{E}^{a1}_{a2}\ \Big(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1p_1}_{i_2i_3} \tag{215}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2} D^{p_1}_{i_3} \tag{216}$$

$$+\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3} D^{p_1}_{i_2}\Big) \tag{217}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} \hat{E}^{a_1i_1p_1}_{i_2i_3a_2} \approx \hat{E}^{a1i1}_{i2a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_3}\right) \tag{218}$$

$$\hat{E}^{a1i1}_{a2i3}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_2}\right) \tag{219}$$

$$\hat{E}^{a1p1}_{i2a2}\ \left(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3}\right) \tag{220}$$

$$\hat{E}^{a1p1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2}\right) \tag{221}$$

$$\hat{E}^{a1}_{a2}\ \Big(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2} D^{p_1}_{i_3} \tag{222}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3} D^{p_1}_{i_2} \tag{223}$$

$$+\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_4i_1} t^{a_3a_1}_{i_3i_2} D^{p_1i_1}_{i_2i_3}\Big) \tag{224}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} \hat{E}^{a_1i_1p_1}_{i_2i_3a_2} \approx \hat{E}^{a1i1}_{i2a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_3}\right) \tag{225}$$

$$\hat{E}^{a1i1}_{a2i3}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_2}\right) \tag{226}$$

$$\hat{E}^{a1p1}_{i2a2}\ \left(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3}\right) \tag{227}$$

$$\hat{E}^{a1p1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2}\right) \tag{228}$$

$$\hat{E}^{a1}_{a2}\ \Big(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2} D^{p_1}_{i_3} \tag{229}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3} D^{p_1}_{i_2} \tag{230}$$

$$+\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_3a_2}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{p_1i_1}_{i_2i_3}\Big) \tag{231}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} \hat{E}^{a_1i_1p_1}_{i_2a_2i_3} \approx \hat{E}^{a1i1}_{i2a2}\ \left(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_3}\right) \tag{232}$$

$$\hat{E}^{a1i1}_{i3a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{p_1}_{i_2}\right) \tag{233}$$

$$\hat{E}^{a1p1}_{i2a2}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3}\right) \tag{234}$$

$$\hat{E}^{a1p1}_{a2i3}\ \left(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2}\right) \tag{235}$$

$$\hat{E}^{a1}_{a2}\ \Big(\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1p_1}_{i_2i_3} \tag{236}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_2} D^{p_1}_{i_3} \tag{237}$$

$$+\tfrac{1}{4}\, f^{i_4}_{p_1} t^{a_2a_3}_{i_1i_4} t^{a_3a_1}_{i_3i_2} D^{i_1}_{i_3} D^{p_1}_{i_2}\Big) \tag{238}$$

the decomposition of the three-body part produces 48 terms; the final block (surviving terms and decomposition merged) has 52 terms:

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right] = -\tfrac{1}{4}\,D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2} \hat{E}^{a_1}_{a_2} \tag{239}$$

$$- \tfrac{1}{4}\,D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1}_{a_2} \tag{240}$$

$$+ \tfrac{1}{2}\,D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1}_{a_2} \tag{241}$$

$$+ \tfrac{1}{2}\,D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1}_{a_2} \tag{242}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{a_2} \tag{243}$$

$$- \tfrac{1}{4}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{a_2} \tag{244}$$

$$- \tfrac{1}{4}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{a_2} \tag{245}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{a_2} \tag{246}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1}_{a_2} \tag{247}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1}_{a_2} \tag{248}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1}_{a_2} \tag{249}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1}_{a_2} \tag{250}$$

$$+ \tfrac{1}{2}\,D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{251}$$

$$- \tfrac{1}{4}\,D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{252}$$

$$+ \tfrac{1}{2}\,D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{253}$$

$$- \tfrac{1}{4}\,D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{254}$$

$$+ \tfrac{1}{2}\,D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1}_{i_1} \tag{255}$$

$$- D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1}_{i_1} \tag{256}$$

$$- \tfrac{1}{4}\,D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1}_{i_1} \tag{257}$$

$$+ \tfrac{1}{2}\,D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1}_{i_1} \tag{258}$$

$$- D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{259}$$

$$+ \tfrac{1}{2}\,D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{260}$$

$$+ \tfrac{1}{2}\,D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{261}$$

$$- \tfrac{1}{4}\,D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{262}$$

$$- \tfrac{1}{4}\,D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{263}$$

$$- \tfrac{1}{4}\,D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1}_{p_1} \tag{264}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{p_1} \tag{265}$$

$$- \tfrac{1}{4}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{p_1} \tag{266}$$

$$- \tfrac{1}{4}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{p_1} \tag{267}$$

$$+ \tfrac{1}{2}\,D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{p_1} \tag{268}$$

$$- \tfrac{1}{2}\,D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{269}$$

$$+ \tfrac{1}{2}\,D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{270}$$

$$+ D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{271}$$
$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{272}$$
$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{273}$$
$$- D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{274}$$
$$+ \tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{275}$$
$$+ \tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{276}$$
$$- \tfrac{1}{2}\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{277}$$
$$+ D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{278}$$
$$+ D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{279}$$
$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{280}$$
$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{281}$$
$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{282}$$
$$+ \tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{283}$$
$$+ \tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{284}$$
$$- D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{285}$$
$$+ \tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{286}$$
$$+ \tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{287}$$
$$+ \tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_2 i_1} t^{a_2 a_3}_{i_4 i_3} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{288}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{289}$$
$$- \tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_1 a_2}_{i_4 i_1} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{290}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (12 operator groups):

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] = \hat{E}^{a1}_{a2} \Big(-\tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2} \tag{291}$$
$$- \tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \tag{292}$$
$$+ \tfrac{1}{2}\, D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{293}$$
$$+ \tfrac{1}{2}\, D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \tag{294}$$
$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \tag{295}$$
$$- \tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \tag{296}$$
$$- \tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \tag{297}$$
$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \tag{298}$$
$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{299}$$
$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \tag{300}$$
$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \tag{301}$$
$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \Big) \tag{302}$$

$$\hat{E}^{a1}_{i1}\ \Big(\tfrac{1}{2}\,D^{i_2i_3}_{i_4p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_4}t^{a_2a_3}_{i_3i_2} \tag{303}$$
$$-\tfrac{1}{4}\,D^{i_2i_3}_{i_4p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_1}t^{a_2a_3}_{i_3i_2} \tag{304}$$
$$+\tfrac{1}{2}\,D^{i_2i_3}_{p_1i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_4}t^{a_2a_3}_{i_2i_3} \tag{305}$$
$$-\tfrac{1}{4}\,D^{i_2i_3}_{p_1i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_1}t^{a_2a_3}_{i_2i_3} \tag{306}$$
$$+\tfrac{1}{2}\,D^{i_2}_{i_3}D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_3}t^{a_2a_3}_{i_2i_4} \tag{307}$$
$$-D^{i_2}_{i_3}D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_3}t^{a_2a_3}_{i_4i_2} \tag{308}$$
$$-\tfrac{1}{4}\,D^{i_2}_{i_3}D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_3i_1}t^{a_2a_3}_{i_2i_4} \tag{309}$$
$$+\tfrac{1}{2}\,D^{i_2}_{i_3}D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_3i_1}t^{a_2a_3}_{i_4i_2} \tag{310}$$
$$-D^{i_2}_{p_1}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_4}t^{a_2a_3}_{i_2i_3} \tag{311}$$
$$+\tfrac{1}{2}\,D^{i_2}_{p_1}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_1i_4}t^{a_2a_3}_{i_3i_2} \tag{312}$$
$$+\tfrac{1}{2}\,D^{i_2}_{p_1}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_1}t^{a_2a_3}_{i_2i_3} \tag{313}$$
$$-\tfrac{1}{4}\,D^{i_2}_{p_1}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_1}t^{a_2a_3}_{i_3i_2}\Big) \tag{314}$$
$$\hat{E}^{a1}_{p1}\ \Big(-\tfrac{1}{4}\,D^{i_1i_2}_{i_3i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_3i_4}t^{a_2a_3}_{i_1i_2} \tag{315}$$
$$-\tfrac{1}{4}\,D^{i_1i_2}_{i_3i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_3}t^{a_2a_3}_{i_2i_1} \tag{316}$$
$$+\tfrac{1}{2}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_4}t^{a_2a_3}_{i_1i_3} \tag{317}$$
$$-\tfrac{1}{4}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_4}t^{a_2a_3}_{i_3i_1} \tag{318}$$
$$-\tfrac{1}{4}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_2}t^{a_2a_3}_{i_1i_3} \tag{319}$$
$$+\tfrac{1}{2}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_2}t^{a_2a_3}_{i_3i_1}\Big) \tag{320}$$
$$\hat{E}^{i1a1}_{a2i2}\ \Big(D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_2i_4}t^{a_2a_3}_{i_1i_3} \tag{321}$$
$$-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_2i_4}t^{a_2a_3}_{i_3i_1} \tag{322}$$
$$-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_4i_2}t^{a_2a_3}_{i_1i_3} \tag{323}$$
$$-D^{i_3}_{p_1}f^{i_4}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_1i_4} \tag{324}$$
$$+\tfrac{1}{2}\,D^{i_3}_{p_1}f^{i_4}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_4i_1} \tag{325}$$
$$+\tfrac{1}{2}\,D^{i_3}_{p_1}f^{i_4}_{p_1}t^{a_1a_3}_{i_3i_2}t^{a_2a_3}_{i_1i_4}\Big) \tag{326}$$
$$\hat{E}^{i1a1}_{i2a2}\ \Big(-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_4i_2}t^{a_2a_3}_{i_3i_1} \tag{327}$$
$$+\tfrac{1}{2}\,D^{i_3}_{p_1}f^{i_4}_{p_1}t^{a_1a_3}_{i_3i_2}t^{a_2a_3}_{i_4i_1}\Big) \tag{328}$$
$$\hat{E}^{i1a1}_{i2p1}\ \Big(-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_2}t^{a_2a_3}_{i_3i_1}\Big) \tag{329}$$
$$\hat{E}^{i1a1}_{i3i2}\ \Big(-\tfrac{1}{2}\,D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_1i_4} \tag{330}$$
$$+D^{i_4}_{p_1}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_4i_1}\Big) \tag{331}$$
$$\hat{E}^{i1a1}_{p1i2}\ \Big(D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_4}t^{a_2a_3}_{i_1i_3} \tag{332}$$
$$-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_2i_4}t^{a_2a_3}_{i_3i_1} \tag{333}$$
$$-\tfrac{1}{2}\,D^{i_3}_{i_4}f^{a_2}_{p_1}t^{a_1a_3}_{i_4i_2}t^{a_2a_3}_{i_1i_3}\Big) \tag{334}$$
$$\hat{E}^{i1i2}_{i3i4}\ \big(\tfrac{1}{2}\,f^{a_1}_{a_2}t^{a_1a_3}_{i_1i_2}t^{a_2a_3}_{i_3i_4} \tag{335}$$

$$+\tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_2 i_1} t^{a_2 a_3}_{i_4 i_3}\Big) \tag{336}$$

$$\hat{E}^{p1a1}_{a2i1}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \tag{337}$$

$$-D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \tag{338}$$

$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2}\Big) \tag{339}$$

$$\hat{E}^{p1a1}_{i1a2}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4}\Big) \tag{340}$$

$$\hat{E}^{p1i1}_{i3i2}\ \big(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \tag{341}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_1 a_2}_{i_4 i_1}\big) \tag{342}$$

2 All final blocks

$\Big[\hat{h},\, \hat{T}_1\Big]$ **(2 terms)**

$$\Big[\hat{h},\, \hat{T}_1\Big] = -h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{343}$$

$\Big[\hat{h},\, \hat{T}_2\Big]$ **(2 terms)**

$$\Big[\hat{h},\, \hat{T}_2\Big] = \tfrac{1}{2}\, h^{a_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{344}$$

$$+\ \tfrac{1}{2}\, h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{345}$$

$\Big[\hat{V},\, \hat{T}_1\Big]$ **(3 terms)**

$$\Big[\hat{V},\, \hat{T}_1\Big] = -g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{346}$$

$$+\ \tfrac{1}{2}\, g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{347}$$

$$+\ \tfrac{1}{2}\, g^{a_1 p_3}_{p_2 p_1} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{p_3 i_1} \tag{348}$$

$\Big[\hat{V},\, \hat{T}_2\Big]$ **(17 terms)**

$$\Big[\hat{V},\, \hat{T}_2\Big] = D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{349}$$

$$-\ \tfrac{1}{2}\, D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{350}$$

$$-\ 2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{351}$$

$$+\ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{352}$$

$$+\ D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{i_1} \tag{353}$$

$$-\ \tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1}_{i_1} \tag{354}$$

$$-\ \tfrac{1}{2}\, D^{i_1 i_2}_{p_2 p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{355}$$

$$-\ \tfrac{1}{2}\, D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{356}$$

$$+ D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{357}$$
$$+ D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{358}$$
$$- \tfrac{1}{2} D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{359}$$
$$- \tfrac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{360}$$
$$+ D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{361}$$
$$- \tfrac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 p_2} \tag{362}$$
$$- \tfrac{1}{2} D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{a_1 p_1}_{p_2 i_1} \tag{363}$$
$$+ \tfrac{1}{4} g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{364}$$
$$+ \tfrac{1}{4} g^{a_1 a_2}_{p_2 p_1} t^{a_1 a_2}_{i_2 i_1} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{365}$$

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right]$ **(2 terms)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_1\right] = - f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{366}$$
$$- f^{a_2}_{i_2} t^{a_2}_{i_1} t^{a_1}_{i_2} \hat{E}^{a_1}_{i_1} \tag{367}$$

$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right]$ **(2 terms)**

$$\left[\left[\hat{F}, \hat{T}_1\right], \hat{T}_2\right] = - \tfrac{1}{2} f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{368}$$
$$- \tfrac{1}{2} f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{369}$$

$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_1\right]$ **(2 terms)**

$$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_1\right] = - \tfrac{1}{2} f^{a_3}_{i_3} t^{a_1 a_3}_{i_1 i_2} t^{a_2}_{i_3} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{370}$$
$$- \tfrac{1}{2} f^{a_3}_{i_3} t^{a_3 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{371}$$

$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_2\right]$ **(0 terms)**

$$\left[\left[\hat{F}, \hat{T}_2\right], \hat{T}_2\right] = 0 \tag{372}$$

$\left[\left[\hat{F}, \hat{T}_1\right]^\dagger, \hat{T}_1\right]$ **(4 terms)**

$$\left[\left[\hat{F}, \hat{T}_1\right]^\dagger, \hat{T}_1\right] = f^{i_1}_{i_2} t^{a_2}_{i_1} t^{a_1}_{i_2} \hat{E}^{a_1}_{a_2} \tag{373}$$
$$- f^{a_2}_{p_1} t^{a_2}_{i_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{374}$$
$$- f^{i_2}_{p_1} t^{a_1}_{i_2} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{375}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(3 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = \tfrac{1}{2}\,f^{a_2}_{a_3}t^{a_2}_{i_1}t^{a_1a_3}_{i_2i_3}\hat{E}^{a_1i_1}_{i_2i_3} \tag{376}$$

$$+\,\tfrac{1}{2}\,f^{a_2}_{a_3}t^{a_2}_{i_1}t^{a_3a_1}_{i_3i_2}\hat{E}^{a_1i_1}_{i_2i_3} \tag{377}$$

$$-\,f^{i_3}_{p_1}t^{a_2a_1}_{i_2i_1}t^{a_2}_{i_3}\hat{E}^{a_1p_1}_{i_1i_2} \tag{378}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(12 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = -\tfrac{1}{2}\,f^{a_3}_{p_1}t^{a_2a_3}_{i_2i_1}t^{a_1}_{i_2}\hat{E}^{a_1i_1}_{a_2p_1} \tag{379}$$

$$-\,\tfrac{1}{2}\,f^{a_3}_{p_1}t^{a_3a_2}_{i_1i_2}t^{a_1}_{i_2}\hat{E}^{a_1i_1}_{a_2p_1} \tag{380}$$

$$-\,\tfrac{1}{2}\,f^{a_3}_{p_1}t^{a_2a_3}_{i_1i_2}t^{a_1}_{i_2}\hat{E}^{a_1i_1}_{p_1a_2} \tag{381}$$

$$-\,\tfrac{1}{2}\,f^{a_3}_{p_1}t^{a_3a_2}_{i_2i_1}t^{a_1}_{i_2}\hat{E}^{a_1i_1}_{p_1a_2} \tag{382}$$

$$+\,\tfrac{1}{2}\,f^{a_2}_{a_3}t^{a_1a_2}_{i_1i_2}t^{a_3}_{i_3}\hat{E}^{i_1i_2}_{a_1i_3} \tag{383}$$

$$+\,\tfrac{1}{2}\,f^{a_2}_{a_3}t^{a_2a_1}_{i_1i_2}t^{a_3}_{i_3}\hat{E}^{i_1i_2}_{i_3a_1} \tag{384}$$

$$+\,\tfrac{1}{2}\,f^{a_1}_{p_1}t^{a_2a_1}_{i_1i_2}t^{a_2}_{i_3}\hat{E}^{i_1i_2}_{i_3p_1} \tag{385}$$

$$+\,\tfrac{1}{2}\,f^{a_1}_{p_1}t^{a_1a_2}_{i_1i_2}t^{a_2}_{i_3}\hat{E}^{i_1i_2}_{p_1i_3} \tag{386}$$

$$-\,\tfrac{1}{2}\,f^{i_3}_{p_1}t^{a_1a_2}_{i_1i_3}t^{a_2}_{i_2}\hat{E}^{i_1p_1}_{a_1i_2} \tag{387}$$

$$-\,\tfrac{1}{2}\,f^{i_3}_{p_1}t^{a_2a_1}_{i_3i_1}t^{a_2}_{i_2}\hat{E}^{i_1p_1}_{a_1i_2} \tag{388}$$

$$-\,\tfrac{1}{2}\,f^{i_3}_{p_1}t^{a_1a_2}_{i_3i_1}t^{a_2}_{i_2}\hat{E}^{i_1p_1}_{i_2a_1} \tag{389}$$

$$-\,\tfrac{1}{2}\,f^{i_3}_{p_1}t^{a_2a_1}_{i_1i_3}t^{a_2}_{i_2}\hat{E}^{i_1p_1}_{i_2a_1} \tag{390}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(52 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = -\tfrac{1}{4}\,D^{i_1i_2}_{i_3i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_3i_4}t^{a_2a_3}_{i_1i_2}\hat{E}^{a_1}_{a_2} \tag{391}$$

$$-\,\tfrac{1}{4}\,D^{i_1i_2}_{i_3i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_4i_3}t^{a_2a_3}_{i_2i_1}\hat{E}^{a_1}_{a_2} \tag{392}$$

$$+\,\tfrac{1}{2}\,D^{i_1p_1}_{i_2i_3}f^{i_4}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_1i_4}\hat{E}^{a_1}_{a_2} \tag{393}$$

$$+\,\tfrac{1}{2}\,D^{i_1p_1}_{i_2i_3}f^{i_4}_{p_1}t^{a_1a_3}_{i_3i_2}t^{a_2a_3}_{i_4i_1}\hat{E}^{a_1}_{a_2} \tag{394}$$

$$+\,\tfrac{1}{2}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_2i_4}t^{a_2a_3}_{i_1i_3}\hat{E}^{a_1}_{a_2} \tag{395}$$

$$-\,\tfrac{1}{4}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_2i_4}t^{a_2a_3}_{i_3i_1}\hat{E}^{a_1}_{a_2} \tag{396}$$

$$-\,\tfrac{1}{4}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_4i_2}t^{a_2a_3}_{i_1i_3}\hat{E}^{a_1}_{a_2} \tag{397}$$

$$+\,\tfrac{1}{2}\,D^{i_1}_{i_2}D^{i_3}_{i_4}f^{a_3}_{a_4}t^{a_1a_4}_{i_4i_2}t^{a_2a_3}_{i_3i_1}\hat{E}^{a_1}_{a_2} \tag{398}$$

$$-\,D^{i_1}_{i_2}D^{i_3}_{p_1}f^{i_4}_{p_1}t^{a_1a_3}_{i_2i_3}t^{a_2a_3}_{i_1i_4}\hat{E}^{a_1}_{a_2} \tag{399}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1}_{a_2} \tag{400}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1}_{a_2} \tag{401}$$

$$- D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1}_{a_2} \tag{402}$$

$$+ \tfrac{1}{2}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{403}$$

$$- \tfrac{1}{4}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{404}$$

$$+ \tfrac{1}{2}\, D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{405}$$

$$- \tfrac{1}{4}\, D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{406}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1}_{i_1} \tag{407}$$

$$- D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1}_{i_1} \tag{408}$$

$$- \tfrac{1}{4}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1}_{i_1} \tag{409}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1}_{i_1} \tag{410}$$

$$- D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{411}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{412}$$

$$+ \tfrac{1}{2}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \hat{E}^{a_1}_{i_1} \tag{413}$$

$$- \tfrac{1}{4}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \hat{E}^{a_1}_{i_1} \tag{414}$$

$$- \tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2} \hat{E}^{a_1}_{p_1} \tag{415}$$

$$- \tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1}_{p_1} \tag{416}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{p_1} \tag{417}$$

$$- \tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{p_1} \tag{418}$$

$$- \tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1}_{p_1} \tag{419}$$

$$+ \tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1}_{p_1} \tag{420}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{421}$$

$$+ \tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{a_2 i_2} \tag{422}$$

$$+ D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{423}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{424}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{425}$$

$$- D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{426}$$

$$+ \tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{427}$$

$$+ \tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 a_2} \tag{428}$$

$$- \tfrac{1}{2}\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{429}$$

$$+ D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{430}$$

$$+ D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{431}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{432}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \hat{E}^{a_1 i_1}_{i_2 p_1} \tag{433}$$

$$- \tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{p_1 i_2} \tag{434}$$

$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1 p_1}_{a_2 i_1} \tag{435}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{436}$$
$$-\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{437}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2} \hat{E}^{a_1 p_1}_{i_1 a_2} \tag{438}$$
$$+\tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{439}$$
$$+\tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_2 i_1} t^{a_2 a_3}_{i_4 i_3} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{440}$$
$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{441}$$
$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_1 a_2}_{i_4 i_1} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{442}$$

3 All final blocks, grouped by operator

Every block with the coefficient pieces collected per operator, ready for substitution.

$\left[\hat{h},\, \hat{T}_1\right]$ **(2 terms, 2 operator groups)**

$$\left[\hat{h},\, \hat{T}_1\right] = \hat{E}^{a1}_{p1}\ \left(-h^{i_1}_{p_1} t^{a_1}_{i_1}\right) \tag{443}$$
$$\hat{E}^{p1}_{i1}\ \left(h^{a_1}_{p_1} t^{a_1}_{i_1}\right) \tag{444}$$

$\left[\hat{h},\, \hat{T}_2\right]$ **(2 terms, 1 operator groups)**

$$\left[\hat{h},\, \hat{T}_2\right] = \hat{E}^{p1a1}_{i2i1}\ \left(\tfrac{1}{2}\, h^{a_2}_{p_1} t^{a_1 a_2}_{i_1 i_2}\right. \tag{445}$$
$$\left.+\tfrac{1}{2}\, h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1}\right) \tag{446}$$

$\left[\hat{V},\, \hat{T}_1\right]$ **(3 terms, 2 operator groups)**

$$\left[\hat{V},\, \hat{T}_1\right] = \hat{E}^{p1a1}_{p3p2}\ \left(-g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1}\right) \tag{447}$$
$$\hat{E}^{p2p1}_{p3i1}\ \left(g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1}\right) \tag{448}$$

$\left[\hat{V},\, \hat{T}_2\right]$ **(17 terms, 6 operator groups)**

$$\left[\hat{V},\, \hat{T}_2\right] = \hat{E}^{a1}_{i1}\ \left(D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2}\right. \tag{449}$$
$$-\tfrac{1}{2}\, D^{i_2 p_1}_{p_2 p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_2 i_1} \tag{450}$$
$$-2\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \tag{451}$$
$$+D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_2 i_1} \tag{452}$$
$$+D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_1 i_2} \tag{453}$$
$$\left.-\tfrac{1}{2}\, D^{i_2}_{p_1} D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_2 i_1}\right) \tag{454}$$

$$\hat{E}^{a1}_{p1}\ \big(-\tfrac{1}{2}\ D^{i_1 i_2}_{p_2 p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_1 a_2}_{i_1 i_2} \tag{455}$$
$$-\tfrac{1}{2}\ D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2} \tag{456}$$
$$+D^{i_1}_{p_2} D^{i_2}_{p_3} g^{a_2 p_1}_{p_3 p_2} t^{a_1 a_2}_{i_1 i_2}\big) \tag{457}$$
$$\hat{E}^{p1a1}_{i1p2}\ \big(-\tfrac{1}{2}\ D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_2 i_1}\big) \tag{458}$$
$$\hat{E}^{p1a1}_{i2i1}\ \big(D^{p_2}_{p_3} g^{a_2 p_3}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \tag{459}$$
$$-\tfrac{1}{2}\ D^{p_2}_{p_3} g^{a_2 p_3}_{p_2 p_1} t^{a_1 a_2}_{i_1 i_2}\big) \tag{460}$$
$$\hat{E}^{p1a1}_{p2i1}\ \big(-\tfrac{1}{2}\ D^{i_2}_{p_3} g^{a_2 p_2}_{p_1 p_3} t^{a_1 a_2}_{i_1 i_2} \tag{461}$$
$$+D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_1 i_2} \tag{462}$$
$$-\tfrac{1}{2}\ D^{i_2}_{p_3} g^{a_2 p_2}_{p_3 p_1} t^{a_1 a_2}_{i_2 i_1}\big) \tag{463}$$
$$\hat{E}^{p1p2}_{i1i2}\ \big(\tfrac{1}{4}\ g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \tag{464}$$
$$+\tfrac{1}{4}\ g^{a_1 a_2}_{p_2 p_1} t^{a_1 a_2}_{i_2 i_1}\big) \tag{465}$$

$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_1\Big]$ **(2 terms, 1 operator groups)**

$$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_1\Big] = \hat{E}^{a1}_{i1}\ \big(-2\,f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1}\big) \tag{466}$$

$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_2\Big]$ **(2 terms, 1 operator groups)**

$$\Big[\Big[\hat{F},\,\hat{T}_1\Big],\,\hat{T}_2\Big] = \hat{E}^{a1a2}_{i1i2}\ \big(-\tfrac{1}{2}\ f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_2 a_3}_{i_2 i_1} \tag{467}$$
$$-\tfrac{1}{2}\ f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2}\big) \tag{468}$$

$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_1\Big]$ **(2 terms, 1 operator groups)**

$$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_1\Big] = \hat{E}^{a1a2}_{i1i2}\ \big(-\tfrac{1}{2}\ f^{a_3}_{i_3} t^{a_1 a_3}_{i_1 i_2} t^{a_2}_{i_3} \tag{469}$$
$$-\tfrac{1}{2}\ f^{a_3}_{i_3} t^{a_3 a_1}_{i_2 i_1} t^{a_2}_{i_3}\big) \tag{470}$$

$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_2\Big]$ **(0 terms, 0 operator groups)**

$$\Big[\Big[\hat{F},\,\hat{T}_2\Big],\,\hat{T}_2\Big] = 0 \tag{471}$$

$\Big[\Big[\hat{F},\,\hat{T}_1\Big]^\dagger,\,\hat{T}_1\Big]$ **(4 terms, 4 operator groups)**

$$\Big[\Big[\hat{F},\,\hat{T}_1\Big]^\dagger,\,\hat{T}_1\Big] = \hat{E}^{a1}_{a2}\ \Big(f^{i_1}_{i_2} t^{a_2}_{i_1} t^{a_1}_{i_2}\Big) \tag{472}$$
$$\hat{E}^{a1}_{p1}\ \big(-f^{a_2}_{p_1} t^{a_2}_{i_1} t^{a_1}_{i_1}\big) \tag{473}$$

$$\hat{E}^{i1}_{i2}\ \left(f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2}\right) \tag{474}$$
$$\hat{E}^{p1}_{i1}\ \left(-f^{i_2}_{p_1} t^{a_1}_{i_2} t^{a_1}_{i_1}\right) \tag{475}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(3 terms, 2 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = \hat{E}^{i1a1}_{i3i2}\ \left(\tfrac{1}{2}\, f^{a_2}_{a_3} t^{a_2}_{i_1} t^{a_1 a_3}_{i_2 i_3}\right. \tag{476}$$
$$\left.+\tfrac{1}{2}\, f^{a_2}_{a_3} t^{a_2}_{i_1} t^{a_3 a_1}_{i_3 i_2}\right) \tag{477}$$
$$\hat{E}^{p1a1}_{i2i1}\ \left(-f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3}\right) \tag{478}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(12 terms, 6 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = \hat{E}^{i1a1}_{a2p1}\ \left(-\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_2 a_3}_{i_1 i_2} t^{a_1}_{i_2}\right. \tag{479}$$
$$\left.-\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_3 a_2}_{i_2 i_1} t^{a_1}_{i_2}\right) \tag{480}$$
$$\hat{E}^{i1a1}_{p1a2}\ \left(-\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_1}_{i_2}\right. \tag{481}$$
$$\left.-\tfrac{1}{2}\, f^{a_3}_{p_1} t^{a_3 a_2}_{i_1 i_2} t^{a_1}_{i_2}\right) \tag{482}$$
$$\hat{E}^{i1i2}_{i3a1}\ \left(\tfrac{1}{2}\, f^{a_2}_{a_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3}\right. \tag{483}$$
$$\left.+\tfrac{1}{2}\, f^{a_2}_{a_3} t^{a_2 a_1}_{i_1 i_2} t^{a_3}_{i_3}\right) \tag{484}$$
$$\hat{E}^{i1i2}_{p1i3}\ \left(\tfrac{1}{2}\, f^{a_1}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_2}_{i_3}\right. \tag{485}$$
$$\left.+\tfrac{1}{2}\, f^{a_1}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3}\right) \tag{486}$$
$$\hat{E}^{p1i1}_{a1i2}\ \left(-\tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_2}_{i_2}\right. \tag{487}$$
$$\left.-\tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2}\right) \tag{488}$$
$$\hat{E}^{p1i1}_{i2a1}\ \left(-\tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_1 a_2}_{i_1 i_3} t^{a_2}_{i_2}\right. \tag{489}$$
$$\left.-\tfrac{1}{2}\, f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2}\right) \tag{490}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(52 terms, 12 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = \hat{E}^{a1}_{a2}\ \left(-\tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2}\right. \tag{491}$$
$$-\tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \tag{492}$$
$$+\tfrac{1}{2}\, D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{493}$$
$$+\tfrac{1}{2}\, D^{i_1 p_1}_{i_2 i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \tag{494}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \tag{495}$$
$$-\tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \tag{496}$$
$$-\tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \tag{497}$$

$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \tag{498}$$
$$-D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{499}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \tag{500}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \tag{501}$$
$$-D^{i_1}_{i_2} D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \Big) \tag{502}$$
$$\hat{E}^{a1}_{i1} \Big(\tfrac{1}{2}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \tag{503}$$
$$-\tfrac{1}{4}\, D^{i_2 i_3}_{i_4 p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \tag{504}$$
$$+\tfrac{1}{2}\, D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \tag{505}$$
$$-\tfrac{1}{4}\, D^{i_2 i_3}_{p_1 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \tag{506}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \tag{507}$$
$$-D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \tag{508}$$
$$-\tfrac{1}{4}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4} \tag{509}$$
$$+\tfrac{1}{2}\, D^{i_2}_{i_3} D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2} \tag{510}$$
$$-D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_2 i_3} \tag{511}$$
$$+\tfrac{1}{2}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_1 i_4} t^{a_2 a_3}_{i_3 i_2} \tag{512}$$
$$+\tfrac{1}{2}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_2 i_3} \tag{513}$$
$$-\tfrac{1}{4}\, D^{i_2}_{p_1} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_1} t^{a_2 a_3}_{i_3 i_2} \Big) \tag{514}$$
$$\hat{E}^{a1}_{p1} \Big(-\tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_3 i_4} t^{a_2 a_3}_{i_1 i_2} \tag{515}$$
$$-\tfrac{1}{4}\, D^{i_1 i_2}_{i_3 i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_3} t^{a_2 a_3}_{i_2 i_1} \tag{516}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \tag{517}$$
$$-\tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \tag{518}$$
$$-\tfrac{1}{4}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \tag{519}$$
$$+\tfrac{1}{2}\, D^{i_1}_{i_2} D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \Big) \tag{520}$$
$$\hat{E}^{i1a1}_{a2i2} \Big(D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \tag{521}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \tag{522}$$
$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3} \tag{523}$$
$$-D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{524}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1} \tag{525}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_1 i_4} \Big) \tag{526}$$
$$\hat{E}^{i1a1}_{i2a2} \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_3}_{a_4} t^{a_1 a_4}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \tag{527}$$
$$+\tfrac{1}{2}\, D^{i_3}_{p_1} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_2} t^{a_2 a_3}_{i_4 i_1} \Big) \tag{528}$$
$$\hat{E}^{i1a1}_{i2p1} \Big(-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_3 i_1} \Big) \tag{529}$$
$$\hat{E}^{i1a1}_{i3i2} \Big(-\tfrac{1}{2}\, D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_1 i_4} \tag{530}$$

$$+D^{i_4}_{p_1} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_3} t^{a_2 a_3}_{i_4 i_1}\big) \tag{531}$$

$$\hat{E}^{i1a1}_{p1i2}\ \Big(D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_1 i_3} \tag{532}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_2 i_4} t^{a_2 a_3}_{i_3 i_1} \tag{533}$$

$$-\tfrac{1}{2}\, D^{i_3}_{i_4} f^{a_2}_{p_1} t^{a_1 a_3}_{i_4 i_2} t^{a_2 a_3}_{i_1 i_3}\Big) \tag{534}$$

$$\hat{E}^{i1i2}_{i3i4}\ \big(\tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \tag{535}$$

$$+\tfrac{1}{2}\, f^{a_1}_{a_2} t^{a_1 a_3}_{i_2 i_1} t^{a_2 a_3}_{i_4 i_3}\big) \tag{536}$$

$$\hat{E}^{p1a1}_{a2i1}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_2 i_4} \tag{537}$$

$$-D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_1 i_3} t^{a_2 a_3}_{i_4 i_2} \tag{538}$$

$$+\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_4 i_2}\Big) \tag{539}$$

$$\hat{E}^{p1a1}_{i1a2}\ \Big(\tfrac{1}{2}\, D^{i_2}_{i_3} f^{i_4}_{p_1} t^{a_1 a_3}_{i_3 i_1} t^{a_2 a_3}_{i_2 i_4}\Big) \tag{540}$$

$$\hat{E}^{p1i1}_{i3i2}\ \big(-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \tag{541}$$

$$-\tfrac{1}{2}\, f^{i_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_1 a_2}_{i_4 i_1}\big) \tag{542}$$